

Kaiyi Guo

Research Interests

As a combination of engineer and designer, my background in **human-computer interaction**, and **wearable sensing** has equipped me with the skills to find and develop the most suitable form factors, sensing modalities, and feature extraction methods for creating more convenient, unobtrusive, and personalized wearable systems tailored for specific people groups.

Education & Experience

Shanghai Jiao Tong University

MS in Computer Science and Technology

Sept 2023 – Jun 2026

(Expected)

- GPA: 3.5/4.0
- *Advisor:* Prof. Dong Wang and Prof. Qian Zhang

University of Michigan, Ann Arbor

Visting scholar in Electrical and Computer Engineering

Sept 2025 – Jan 2026

- *Advisor:* Prof. Junyi Zhu
- Develop an Electrical Impedance Tomography (EIT) project aimed at monitoring specific patient populations

Zhengzhou University

BS in Software Engineering

Sept 2019 – Jun 2023

- GPA: 3.8/4.0 (rank: 1/58)
- **Coursework:** Computer Architecture, Computer Network, Operate System

Publications

- C5.** Cong Liu*, Ruihao Zheng*, Junbin Ren, **Kaiyi Guo**, Qian Zhang, Dong She, Yuting Bai, Zhanpeng Jin, Yang Gao, **RageSense: Leveraging Acoustic Sensing and LLM-Based Intervention for Privacy-Preserving Emotion Regulation in Mobile Gaming**, In Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26).
- J2.** Ahsan Jamal Akbar, **Kaiyi Guo**, Qian Zhang, Dong Wang, **EchoLip: Pushing the Limit of Acoustic-Based Silent Speech Interface on Mobile Devices**, In Proceedings of IEEE Internet of Things Journal (IEEE IOTJ).
- C4.** **Kaiyi Guo***, Tianyu Wu*, Yang Gao, Qian Zhang, Dong Wang, **EchoTouch: Low-power Face-touching Behavior Recognition Using Active Acoustic Sensing on Glasses**, In Proceedings of ACM Interact. Mob. Wearable Ubiquitous Technol (IMWUT)/Ubicomp'25.
- J1.** **Kaiyi Guo**, Qian Zhang, Dong Wang, **EchoExpress: Facial Expression Recognition in the Wild via Acoustic Sensing on Smart Glasses**, In Proceedings of IEEE Transactions on Mobile Computing (IEEE TMC).
- C3.** **Kaiyi Guo**, Qian Zhang, Dong Wang, **EchoBreath: Continuous Respiratory Behavior Recognition in the Wild via Acoustic Sensing on Smart Glasses**, In Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI'25, **Honorable Mention**).
- C2.** Qian Zhang, **Kaiyi Guo**, Yifei yang, Dong Wang, **WearSE: Enabling Streaming Speech Enhancement on Eyewear Using Acoustic Sensing**, In Proceedings of ACM Interact. Mob. Wearable Ubiquitous Technol (IMWUT)/Ubicomp'25.
- C1.** Qian Zhang, Yubin Lan, **Kaiyi Guo**, Dong Wang, **Lipwatch: Enabling Silent Speech Recognition on Smartwatches using Acoustic Sensing**, In Proceedings of ACM Interact. Mob. Wearable Ubiquitous Technol (IMWUT)/Ubicomp'24.

* Equal Contribution

Awards

CHI 2025 Honorable Mention
ACM SIGCHI

Outstanding Graduate
Shanghai Jiao Tong University

Service

External Reviewer

2025 – present

*IMWUT'25,26 (Outstanding reviewer),
UIST'25 (Outstanding reviewer),
CHI'26,
IEEE TMC'25,26,
DIS'26*

Projects

Acoustic Sensing Based Sleep Monitoring System

Lab Project

- Develop a smart speaker system that emits ultrasound waves and analyzes the received echo profiles to obtain critical sleep health indicators, such as sleep stages and sleep apnea. Design and implement a cloud-based Python backend for data processing and storage, and create a user-friendly web interface for visualizing sleep stages, apnea events, and personalized sleep insights.
- Tools Used: C++, Python

Teaching Experience

SE3303 Data Mining and Big Data Analytics

Spring'25

*Teaching Assistant at Shanghai Jiao Tong University, Instructor: Prof. Dong Wang
and Prof. Qian Zhang*

EI8702 Internet of Things (IoT) Technology

Fall'24

*Teaching Assistant at Shanghai Jiao Tong University, Instructor: Prof. Dong Wang
and Prof. Qian Zhang*

SE3303 Data Mining and Big Data Analytics

Spring'24

*Teaching Assistant at Shanghai Jiao Tong University, Instructor: Prof. Dong Wang
and Prof. Qian Zhang*

Skills

Languages: C++, C, Python, Java

Technologies: Tensorflow, Pytorch